MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 12

1. *Let* $\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\\1\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$ *and* $\mathbf{v}_3 = \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}$. *Using Gram-Schmidt, find an orthonormal basis for* $W = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, *using* $\mathbf{v}_1, \mathbf{v}_2,$ *and* $\mathbf{v}_3$.

Set

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\begin{bmatrix}1\\0\\1\\1\end{bmatrix}}{\left\|\begin{bmatrix}1\\0\\1\\1\end{bmatrix}\right\|} = \begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix}$$

Then

$$\mathbf{u}_2 = \frac{\mathbf{v}_2 - (\mathbf{u}_1 \cdot \mathbf{v}_2)\mathbf{u}_1}{\|\mathbf{v}_2 - (\mathbf{u}_1 \cdot \mathbf{v}_2)\mathbf{u}_1\|} = \frac{\begin{bmatrix}1\\0\\0\\0\end{bmatrix} - \left(\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix} \cdot \begin{bmatrix}1\\0\\0\\0\end{bmatrix}\right)\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix}}{\left\|\begin{bmatrix}1\\0\\0\\0\end{bmatrix} - \left(\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix} \cdot \begin{bmatrix}1\\0\\0\\0\end{bmatrix}\right)\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix}\right\|} = \frac{\begin{bmatrix}\frac{2}{3}\\0\\-\frac{1}{3}\\-\frac{1}{3}\end{bmatrix}}{\left\|\begin{bmatrix}\frac{2}{3}\\0\\-\frac{1}{3}\\-\frac{1}{3}\end{bmatrix}\right\|} = \sqrt{\frac{3}{2}}\begin{bmatrix}\frac{2}{3}\\0\\-\frac{1}{3}\\-\frac{1}{3}\end{bmatrix} = \begin{bmatrix}\frac{2}{\sqrt{6}}\\0\\-\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\end{bmatrix}$$

Finally,

$$\mathbf{u}_3 = \frac{\mathbf{v}_3 - (\mathbf{u}_1 \cdot \mathbf{v}_3)\,\mathbf{u}_1 - (\mathbf{u}_2 \cdot \mathbf{v}_3)\mathbf{u}_2}{\|\mathbf{v}_3 - (\mathbf{u}_1 \cdot \mathbf{v}_3)\mathbf{u}_1 - (\mathbf{u}_2 \cdot \mathbf{v}_3)\mathbf{u}_2\|}$$

$$= \frac{\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} - \left(\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix} \cdot \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}\right)\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix} - \left(\begin{bmatrix}\frac{2}{\sqrt{6}}\\0\\-\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\end{bmatrix} \cdot \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}\right)\begin{bmatrix}\frac{2}{\sqrt{6}}\\0\\-\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\end{bmatrix}}{\left\|\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} - \left(\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix} \cdot \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}\right)\begin{bmatrix}\frac{1}{\sqrt{3}}\\0\\\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{3}}\end{bmatrix} - \left(\begin{bmatrix}\frac{2}{\sqrt{6}}\\0\\-\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\end{bmatrix} \cdot \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}\right)\begin{bmatrix}\frac{2}{\sqrt{6}}\\0\\-\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\end{bmatrix}\right\|}$$

$$= \sqrt{\frac{2}{3}}\begin{bmatrix}0\\1\\\frac{1}{2}\\-\frac{1}{2}\end{bmatrix} = \begin{bmatrix}0\\\frac{2}{\sqrt{6}}\\\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\end{bmatrix}$$

Now $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis of W.

2. *Previously, we learned that given a basis of some subspace W, the Gram-Schmidt process produces an* orthonormal *basis of the same subspace W. In fact, we need not start with a basis—it is sufficient to start with any spanning set of W, even if the set is not linearly independent.*

Let $\mathbf{v}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}2\\1\\1\end{bmatrix}$. *Apply Gram-Schmidt to* $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ *and obtain an orthonormal basis for the subspace* $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

To get the first vector, we normalize $\mathbf{v}_1$.

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix}.$$

Next, we calculate $\mathbf{b}_2$ and normalize it to get $\mathbf{u}_2$.

$$\mathbf{b}_2 = \begin{bmatrix}1\\0\\0\end{bmatrix} - \left(\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix} \cdot \begin{bmatrix}1\\0\\0\end{bmatrix}\right)\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}\frac{2}{3}\\-\frac{1}{3}\\-\frac{1}{3}\end{bmatrix}.$$

Therefore $\mathbf{u}_2 = \frac{1}{\sqrt{6}}\begin{bmatrix}2\\-1\\-1\end{bmatrix}$.

Finally we solve for $\mathbf{b}_3$ and $\mathbf{u}_3$.

$$\mathbf{b}_3 = \begin{bmatrix}2\\1\\1\end{bmatrix} - \left(\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix} \cdot \begin{bmatrix}2\\1\\1\end{bmatrix}\right)\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix} - \left(\frac{1}{\sqrt{6}}\begin{bmatrix}2\\-1\\-1\end{bmatrix} \cdot \begin{bmatrix}2\\1\\1\end{bmatrix}\right)\frac{1}{\sqrt{6}}\begin{bmatrix}2\\-1\\-1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}.$$

So the vector $\mathbf{u}_3$ does not exist! This means that we can discard the last vector to get a basis.

That is, our orthonormal basis is $\{\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix}2\\-1\\-1\end{bmatrix}\}$.

3. *Compute the QR-factorization for each of the following matrices (you may assume the columns are linearly independent):*

(1) $\begin{bmatrix}1 & 2 & 5\\-1 & 1 & -4\\-1 & 4 & -3\\1 & -4 & 7\\1 & 2 & 1\end{bmatrix}$

(2) $\begin{bmatrix}1 & 1 & 1\\1 & 1 & 0\\1 & 0 & 0\\1 & 0 & 0\end{bmatrix}$

(3) $\begin{bmatrix}1 & 1 & 1\\0 & 1 & 1\\0 & 0 & 1\\0 & 0 & 1\end{bmatrix}$

In each of the parts, the matrix Q is the same size as A. The columns of Q are obtained by applying the Gram-Schmidt algorithm to the columns of A (of course, keeping the columns of A in the same order as they appear in A). Then R is obtained by $R = Q^TA$.

(1)
$$\begin{bmatrix}1 & 2 & 5\\-1 & 1 & -4\\-1 & 4 & -3\\1 & -4 & 7\\1 & 2 & 1\end{bmatrix} = \underbrace{\begin{bmatrix}1/\sqrt{5} & 1/2 & 1/2\\-1/\sqrt{5} & 0 & 0\\-1/\sqrt{5} & 1/2 & 1/2\\1/\sqrt{5} & -1/2 & 1/2\\1/\sqrt{5} & 1/2 & -1/2\end{bmatrix}}_{Q} \underbrace{\begin{bmatrix}\sqrt{5} & -\sqrt{5} & 4\sqrt{5}\\0 & 6 & -2\\0 & 0 & 4\end{bmatrix}}_{R}$$

(2)
$$\begin{bmatrix}1 & 1 & 1\\1 & 1 & 0\\1 & 0 & 0\\1 & 0 & 0\end{bmatrix} = \begin{bmatrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}}\\\frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}}\\\frac{1}{2} & -\frac{1}{2} & 0\\\frac{1}{2} & -\frac{1}{2} & 0\end{bmatrix}\begin{bmatrix}2 & 1 & \frac{1}{2}\\0 & 1 & \frac{1}{2}\\0 & 0 & \frac{1}{\sqrt{2}}\end{bmatrix}$$

(3)
$$\begin{bmatrix}1 & 1 & 1\\0 & 1 & 1\\0 & 0 & 1\\0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & \frac{1}{\sqrt{2}}\\0 & 0 & \frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\0 & 1 & 1\\0 & 0 & \sqrt{2}\end{bmatrix}$$

4. *Let* $A = \begin{bmatrix}1 & 0\\0 & 1\\1 & 1\end{bmatrix}$.

a. *Find the QR decomposition of A: write $A = QR$ where Q is a matrix with orthonormal columns and R is an upper triangular matrix.*

b. *Let* $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. *Use the QR decomposition of A to find the least squares solution of* $A\hat{\mathbf{x}} = \mathbf{b}$ *(by solving* $R\hat{\mathbf{x}} = Q^T\mathbf{b}$*).*

a. We start with columns of $A(= [\mathbf{v}_1\ \mathbf{v}_2])$ and we use Gram-Schmidt to find columns of $Q(= [\mathbf{q}_1\ \mathbf{q}_2])$:

$$q_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

and,

$$\mathbf{q}_2 = \frac{\mathbf{v}_2 - (\mathbf{q}_1 \cdot \mathbf{v}_2)\mathbf{q}_1}{\|\mathbf{v}_2 - (\mathbf{q}_1 \cdot \mathbf{v}_2)\mathbf{q}_1\|} = \frac{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \left(\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right) \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}}{\left\| \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \left(\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right) \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\|} = \frac{\begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix}}{\left\| \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} \right\|} = \sqrt{\frac{2}{3}} \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

Hence,

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

We have:

$$R = Q^T A = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} \end{bmatrix}$$

b. We have to solve $R\hat{\mathbf{x}} = Q^T\mathbf{b}$:

$$\begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

Therefore, $\hat{\mathbf{x}} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$.

5. *Let* $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

(1) *Find a diagonal matrix D and an invertible matrix P such that* $A = PDP^{-1}$.
(2) *Can you choose P to an be orthogonal matrix?*

We have to find the eigenvalues and corresponding eigenspaces. We have:

$$\det \begin{bmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix} = (1-\lambda)(1-\lambda) - 1 = -\lambda(2-\lambda)$$

Hence, the eigenvalues of A are 0 and 2. For $\lambda = 0$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R2 \to R2 - R1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Hence, the corresponding eigenspace is span $\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$.

For $\lambda = 2$:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{R2 \to R2 + R1, R1 \to -R1} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

Hence, the corresponding eigenspace is span $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$.

Columns of P are (linearly independent) eigenvectors of A and D is the diagonal matrix with eigenvalues of A on the main diagonal in the appropriate order(corresonding to columns of P). Therefore:

$$P = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \qquad D = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

6. *True/False. Give reasons why!*

(1) *The QR decomposition of a matrix is unique.*
(2) *If $A = QR$, where Q has orthonormal columns, then $R = Q^T A$.*
(3) *In a QR factorization, say $A = QR$ (when A has linearly independent columns), the columns of Q form an orthonormal basis for the column space of A.*
(4) *If $A = QR$ (as in the QR decomposition), then Col(A)= Col (R).*

(1) True. If A is invertible, then the factorization is unique if we require that the diagonal elements of R be positive.
(2) True, because the $Q^T Q = I$.
(3) True. Let Qx be a vector in the column space of Q. Let $y := R^{-1}x$. Then $Qx = QRy = Ay$ is in the column space of A. Thus every vector in the column space of Q is in the column space of A. Thus the columns of Q are in the column space of A and thus form an orthonormal basis of the column space of A.
(4) False. Take for example the QR decomposition in Problem 1(1). There A is a 4×4-matrix and R is a 3×3-matrix. Thus there column spaces can not be equal.